LESSON 5.5

Determine the Volume of Rectangular Prisms

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the volume of a rectangular prism with whole-number edges using length × width × height.

WHY THIS MATTERS Your team is building time capsule containers to store class memories. Each container is a rectangular prism, and you need to figure out how much each one can hold. That means calculating the volume using whole number dimensions! Container A is 5 inches long, 3 inches wide, and 4 inches tall. How much can it hold?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Volume *Spanish: Volumen*	How much space is inside a solid shape.	V = l × w × h
Rectangular prism *Spanish: Prisma rectangular*	A solid box shape with six flat rectangle sides.	a cereal box
Cubic units *Spanish: Unidades cúbicas*	The units used to measure space inside, like cubic inches.	cm³, in³, ft³
Length, width, height *Spanish: Largo, ancho, altura*	How long, how wide, and how tall a box is.	V = l × w × h: for a box 5 × 3 × 2, volume = 30 cubic units
Net *Spanish: Plantilla (desarrollo plano)*	A flat shape that folds up into a solid.	A cross-shaped pattern of 6 rectangles folds into a rectangular box
Edge *Spanish: Arista*	The line where two flat sides of a solid meet.	A cube has 12 edges — 4 along the top, 4 along the bottom, 4 vertical

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A box is 6 cm long, 3 cm wide, and 4 cm tall.

1. One layer: 6 × 3 = 18 cubes.
2. 4 layers stack up.
3. 18 × 4 = 72.

Answer: V = 72 cm³

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller box: 5 long, 2 wide, 2 tall. What formula do we use?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The length is _____ , the width is _____ , and the height is _____ .
- The base area is _____ × _____ = _____ square units.
- The volume is _____ cubic units.
- I used cubic units instead of square units because _____ .

Word bank volume | rectangular prism | length | width | height | base area | layer | unit cube | cubic units | multiply | solid | face

WATCH OUT!

A common mistake

A common mistake in Volume with Whole Number Edges is multiplying only two of the three edges and stopping there

LESSON 5.5

Determine the Volume of Rectangular Prisms

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Your team is building time capsule containers to store class memories. Each container is a rectangular prism, and you need to figure out how much each one can hold. That means calculating the volume using whole number dimensions! Container A is 5 inches long, 3 inches wide, and 4 inches tall. How much can it hold?

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the volume of each rectangular prism.

Length	Width	Height	Volume
5 in	3 in	4 in	
6 cm	6 cm	2 cm	
9 ft	2 ft	5 ft	

HINT Each row gives a length, width, and height. The empty Volume cell wants one number with cubic units.

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Put these steps in order for finding the volume of a box that is 8 in long, 5 in wide, and 3 in tall.

Groups:

_____ Identify: l = 8, w = 5, h = 3

_____ Write the formula: V = l × w × h

_____ Multiply: 8 × 5 = 40

_____ Multiply: 40 × 3 = 120

_____ Answer: V = 120 in³

HINT Read each step card. One identifies numbers, one writes a formula, two multiply, and one states the answer.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

What is the volume of a rectangular prism with l = 7 in, w = 3 in, h = 4 in?

(A) 14 in³
(B) 42 in³
(C) 84 in³
(D) 84 in²

HINT Notice the three edge lengths: l = 7 in, w = 3 in, h = 4 in. Volume asks how much space fills the whole box.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

A time capsule box has a volume of 120 cm³. Its length is 10 cm and width is 4 cm. What is its height?

(A) 3 cm
(B) 4 cm
(C) 6 cm
(D) 12 cm

HINT You already know the volume (120 cm³) and two edges (10 cm and 4 cm). The height is the missing piece.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

Which unit is used for volume?

(A) Square inches (in²)
(B) Cubic inches (in³)
(C) Inches (in)
(D) Degrees (°)

HINT Re-read each unit and think dimensions: volume fills 3D space — length, width, AND height.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

The Forgotten Height

1. **Dimensions** A time capsule box is 4 long, 3 wide, and 5 tall
2. **Formula** V = length × width × height
3. **Calculate** V = 4 × 3 = 12 cubic units
4. **Answer** Volume = 12 cubic units

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Look at the Calculate step. How many edges did the student multiply?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Container C holds the most (96 in³) and Container B holds the least (50 in³). What do you notice about the dimensions that make one volume larger than another?

SENTENCE STARTER Container C has the largest volume because ___ × ___ × ___ = ___. Compared to Container B, the dimensions ___ produce a ___ volume because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.5

Determine the Volume of Rectangular Prisms

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the volume of a rectangular prism with whole-number edges using length × width × height.

WHY THIS MATTERS Your team is building time capsule containers to store class memories. Each container is a rectangular prism, and you need to figure out how much each one can hold. That means calculating the volume using whole number dimensions! Container A is 5 inches long, 3 inches wide, and 4 inches tall. How much can it hold?

PART 1 Understand the idea

models and meaning

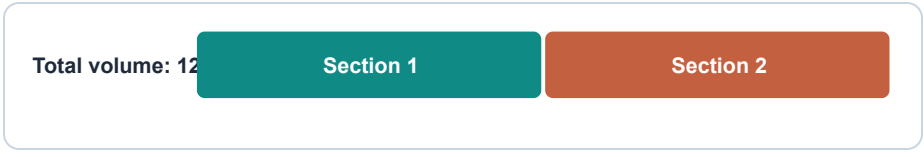
1 **Complete the table. Show how you found each value.**

Complete the table. Find the missing dimension or volume for each time capsule.

Capsule	Length	Width	Height	Volume
Capsule X	9 in	5 in	2 in	
Capsule Y	6 ft	4 ft		168 ft³
Capsule Z	12 cm		5 cm	360 cm³

SHOW YOUR WORK

2 **Use the model to solve. Show your work.**



Each section holds how many cubic inches?

SHOW YOUR WORK

Answer: _____

3 **Write the name of the correct group on each line.**

Sort these rectangular prisms from LEAST volume to GREATEST volume.

Groups:

_____ 3 × 3 × 3

_____ 10 × 2 × 2

_____ 5 × 4 × 3

_____ 6 × 6 × 2

_____ 8 × 5 × 4

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Two storage bins: Bin A is 8 × 6 × 5 inches. Bin B is 9 × 4 × 7 inches. Which holds more and by how much?

(A) Bin B by 12 in³

(B) Bin A by 12 in³

(C) Bin B by 24 in³

(D) They hold the same

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A rectangular prism has length 8 cm, width 5 cm, and height 4 cm. What is its volume?

(A) 17 cm³

(B) 80 cm³

(C) 120 cm³

(D) 160 cm³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Added the Last Edge

1. **Dimensions** A capsule is 6 long, 5 wide, and 3 tall
2. **Formula** V = length × width × height
3. **Calculate** V = 6 × 5 + 3 = 33 cubic units
4. **Answer** Volume = 33 cubic units

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Container C holds the most (96 in³) and Container B holds the least (50 in³). What do you notice about the dimensions that make one volume larger than another?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.5

Determine the Volume of Rectangular Prisms

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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WHY THIS MATTERS Your team is building time capsule containers to store class memories. Each container is a rectangular prism, and you need to figure out how much each one can hold. That means calculating the volume using whole number dimensions! Container A is 5 inches long, 3 inches wide, and 4 inches tall. How much can it hold?

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

If you double ONE dimension of a 3 × 4 × 5 box, how does the volume change?

Change	New Dimensions	New Volume	How did it change?
Double the length	6 × 4 × 5		
Double the width	3 × 8 × 5		
Double the height	3 × 4 × 10		

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A shipping crate is 12 ft long, 3 ft wide, and 2 ft tall. How many 1-foot cubes exactly fill it?

(A) 17 cubes
(B) 36 cubes
(C) 72 cubes
(D) 144 cubes

SHOW YOUR WORK

3 Circle the letter of the best answer. Show how you know.

Two prisms have the same volume of 48 cm³. Prism A is 4 × 4 × 3. Prism B is 8 × 3 × h. What is h?

(A) 2 cm
(B) 3 cm
(C) 4 cm
(D) 6 cm

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

PART 3 Explain and correct

reasoning

4 Find the mistake. Explain it, then show the correct work.

Find the Volume Error

1. **Dimensions** l = 9 ft, w = 4 ft, h = 6 ft
2. **Formula** V = l × w × h
3. **Calculate** V = 9 × 4 × 6 = 9 × 10 = 90 ft³
4. **Answer** Volume = 90 ft³

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

5 **Answer in complete sentences.**

A box needs to hold exactly 60 cubic inches. Give three different sets of whole-number dimensions that work. Which set would make the box closest to a cube shape? Why might that matter?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Container C holds the most (96 in³) and Container B holds the least (50 in³). What do you notice about the dimensions that make one volume larger than another?

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